Compositional Systems in Finite-Dimensional Classical Mechanics

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Abstract

We give a typed formulation of finite-dimensional classical mechanics in which systems, states, observables, processes, and composition are distinct data. Smooth systems and smooth processes form a category, and Cartesian product supplies a symmetric monoidal operation. For symplectic systems we prove that the product two-form is symplectic and that additive Hamiltonians generate factorized dynamics. Interaction terms preserve kinematical product structure while generally destroying dynamical factorization. Counterexamples show that a product state space does not imply independence, that arbitrary smooth maps are not Hamiltonian processes, that reduced spaces need not be smooth manifolds, and that mathematical isomorphism does not establish physical equivalence. Empirical realization is therefore represented as additional, fallible structure rather than as a consequence of the mathematics. Every substantive assertion is assigned an epistemic type.

1 Scope and epistemic discipline

The subject is finite-dimensional smooth classical mechanics. Field theories, continuum mechanics, general relativity, quantum mechanics, and infinite-dimensional phase spaces enter only as boundary cases. Standard symplectic mechanics is documented in Abraham and Marsden [1] and Marsden and Ratiu [2]. Category-theoretic terminology follows Mac Lane [3]; the use of symmetric monoidal categories as a language of processes is discussed by Baez and Stay [4]. These references support established mathematics and physical formalism, not the empirical adequacy of any particular model.

Assumption 1.1 (Finite-dimensional regularity, A1)**.** *Every state space considered in the main construction is a finite-dimensional, Hausdorff, second-countable smooth manifold, and every displayed observable or process has the differentiability stated for it.*

Interpretation 1.2 (Founding thesis under investigation, I1)**.** *The sentence "physics studies empirically realized compositional mathematical structures" is treated as a methodological proposal. It is neither an axiom of mathematics nor an established physical result. The construction below tests whether its terms can be separated precisely in classical mechanics.*

A claim labelled Definition fixes terminology. An Axiom stipulates formal closure. An Assumption restricts the model class. Propositions, Lemmas, Theorems, and Corollaries are proved. An Interpretation connects mathematics to possible physical meaning without asserting experimental fact. An Empirical statement identifies input that must be tested. An Established physical result records accepted physical formalism supported by the cited literature. A New result records a synthesis specific to this work. A Conjecture remains unproved.

2 Systems, states, and observables

Definition 2.1 (Kinematical system, D1)**.** *A* kinematical classical system *is a pair* $S = (X, \mathcal{O}_S)$*, where* X *is a manifold and* $\mathcal{O}_S \subseteq C^\infty(X, \mathbb{R})$ *is a unital real subalgebra separating the states the model intends to distinguish. A state is a point* $x \in X$*, and an observable is an element* $f \in \mathcal{O}_S$*.*

The separation clause belongs to the mathematical model. It does not assert that an experiment resolves every pair of represented states.

Definition 2.2 (Hamiltonian system, D2)**.** *A* Hamiltonian system *is a quadruple* $(X, \omega, \mathcal{O}_S, h)$*, where* (X, ω) *is symplectic,* $\mathcal{O}_S$ *is as in Definition 2.1, and* $h \in C^\infty(X, \mathbb{R})$*. Symplectic means that* $d\omega = 0$ *and that* $v \mapsto \omega_x(v, \cdot)$ *is injective for every* x*.*

Nondegeneracy supplies a unique vector field X_h satisfying

$$\iota_{X_h}\omega = dh. \tag{1}$$

Define $\{f, g\} = \omega(X_f, X_g)$. Along any integral curve γ of X_h,

$$\frac{d}{dt} f(\gamma(t)) = df(X_h) = \{f, h\}.$$

This equality is a calculation conditional on the existence of the integral curve.

Established physical result 2.3 (Hamiltonian representation, E1)**.** *Finite-dimensional conservative classical models are standardly represented by symplectic state spaces, Hamiltonian functions, and Hamiltonian flows [2]. This established formalism does not assert that every physical system is finite-dimensional, conservative, or Hamiltonian.*

Example 2.4 (Particle on a line)**.** *On* $T^*\mathbb{R}$ *with coordinates* (q, p) *and* $\omega = dq \wedge dp$*, the Hamiltonian* $h = p^2/(2m) + V(q)$ *gives* $\dot{q} = p/m$ *and* $\dot{p} = -V'(q)$ *by Equation* (1)*.*

Counterexample 2.5 (Observables need not separate states)**.** *If* $X = \mathbb{R}$ *and* $\mathcal{O}$ *consists only of constant functions, distinct points give equal values for every declared observable. A manifold alone therefore does not determine an adequate observable algebra.*

3 Processes and their category

Definition 3.1 (Typed process, D3)**.** *For* $S = (X, \mathcal{O}_S)$ *and* $T = (Y, \mathcal{O}_T)$*, a process* $F : S \to T$ *is a smooth map* $F : X \to Y$ *satisfying* $F^*(\mathcal{O}_T) \subseteq \mathcal{O}_S$*. Its source and target are part of its type.*

Axiom 3.2 (Process closure, A2)**.** *Every identity is admissible, and the composite of composable admissible processes is admissible.*

Proposition 3.3 (Category of systems, P1)**.** *Kinematical systems and typed processes form a category* $\mathsf{SmoothSys}$*.*

Proof. Axiom A2 supplies identities and closure. Function composition is associative and satisfies both identity laws. If $F : S \to T$, $G : T \to U$, and $u \in \mathcal{O}_U$, then $(G \circ F)^* u = F^*(G^* u) \in \mathcal{O}_S$. Thus composition retains the observable condition. □

Definition 3.4 (Hamiltonian-preserving process, D4)**.** *A diffeomorphism* $F : (X, \omega_X, h_X) \to (Y, \omega_Y, h_Y)$ *is Hamiltonian-preserving when* $F^*\omega_Y = \omega_X$ *and* $h_Y \circ F = h_X$*.*

Proposition 3.5 (Intertwining dynamics, P2). *A Hamiltonian-preserving process satisfies* $TF \circ X_{h_X} = X_{h_Y} \circ F$ *and therefore intertwines the local flows wherever they are defined.*

Proof. For $v \in T_x X$,

$$\omega_Y(TF(X_{h_X}), TF(v)) = \omega_X(X_{h_X}, v) = dh_X(v) = dh_Y(TF(v)).$$

Surjectivity of TF and nondegeneracy of ω_Y identify the vector fields. Uniqueness for smooth ordinary differential equations gives the flow statement. □

Counterexample 3.6 (Smooth is not symplectic). *On* $(\mathbb{R}^2, dq \wedge dp)$, $F(q,p) = (2q, 2p)$ *satisfies* $F^*(dq \wedge dp) = 4dq \wedge dp$. *It is a smooth diffeomorphism but not a symplectic process.*

4 Parallel composition

Definition 4.1 (Parallel composite, D5). *Set* $S \otimes T = (X \times Y, \mathcal{O}_{S\otimes T})$, *where* $\mathcal{O}_{S\otimes T}$ *is generated by* $\pi_X^* \mathcal{O}_S$ *and* $\pi_Y^* \mathcal{O}_T$. *Set* $F \otimes G = F \times G$. *The unit is the one-point system.*

Proposition 4.2 (Symmetric monoidal kinematics, P3). *Cartesian product makes* SmoothSys *symmetric monoidal up to the canonical product diffeomorphisms.*

Proof. Product maps preserve identities and composition pointwise. The associator, unitors, and swap map are diffeomorphisms preserving the generated observable algebras. The pentagon, triangle, and hexagon equations hold because their two sides send each tuple to the same rebracketing or permutation [3]. □

Lemma 4.3 (Product symplectic form, L1). *If* (X, ω_X) *and* (Y, ω_Y) *are symplectic, then* $\omega = \pi_X^* \omega_X + \pi_Y^* \omega_Y$ *is symplectic on* $X \times Y$.

Proof. Closedness follows from commutation of exterior differentiation with pullback. If (u, v) is in the kernel, pairing it with $(u', 0)$ gives $u = 0$ by nondegeneracy of ω_X; pairing with $(0, v')$ then gives $v = 0$. □

Theorem 4.4 (Additive dynamics, T1). *For* $h_0 = \pi_X^* h_X + \pi_Y^* h_Y$,

$$X_{h_0}(x, y) = (X_{h_X}(x), X_{h_Y}(y)).$$

Proof. For $(u, v) \in T_x X \oplus T_y Y$,

$$\omega((X_{h_X}, X_{h_Y}), (u, v)) = dh_X(u) + dh_Y(v) = dh_0(u, v).$$

Uniqueness in Equation (1) proves the formula. □

Corollary 4.5 (Factorized flow, C1). *Where the component flows exist,* $\phi_t^{h_0}(x, y) = (\phi_t^{h_X}(x), \phi_t^{h_Y}(y))$.

Proof. The product curve has derivative X_{h_0} by Theorem T1 and has the prescribed initial point. Uniqueness gives the equality. □

Counterexample 4.6 (Product does not imply independence). *For two oscillators with interaction* $V(q_1, q_2) = kq_1q_2$, $k \neq 0$, *the equation* $\dot{p}_1 = -m_1\omega_1^2 q_1 - kq_2$ *depends on the second subsystem. The product state manifold therefore neither states statistical independence nor entails dynamically independent evolution.*

Counterexample 4.7 (Completeness is not automatic). *For* $h(q,p) = q^2 p$ *on* $\mathbb{R}^2$, *Hamilton's equation* $\dot{q} = q^2$ *has solution* $q(t) = q_0/(1 - q_0 t)$ *for* $q_0 > 0$. *It diverges at* $t = 1/q_0$. *Smooth Hamiltonian data need not generate global time-indexed processes.*

4.1 Observable embeddings and coupled evolution

The product construction admits each subsystem observable without identifying it with an observable of the other subsystem. Concretely, the maps $j_X(f) = f \circ \pi_X$ and $j_Y(g) = g \circ \pi_Y$ are injective unital algebra homomorphisms whenever the factor observable algebras separate their declared states. For the product symplectic form they also satisfy

$$\{j_X(f), j_Y(g)\} = 0.$$

This vanishing is kinematical: it follows from the complementary tangent factors occupied by the two Hamiltonian vector fields. It should not be confused with constancy of either observable during an interacting evolution.

Proposition 4.8 (Interaction test, P5)**.** *Let $h = h_0 + V$ on $X \times Y$, with h_0 as in Theorem T1. For $f \in C^\infty(X)$, the evolution of $j_X(f)$ is*

$$\frac{d}{dt} j_X(f) = j_X(\{f, h_X\}_X) + \{j_X(f), V\}.$$

Thus the first-factor observable follows its isolated equation for every f if and only if dV annihilates every vector of the form $(X_f, 0)$.

Proof. The displayed identity is bilinearity of the Poisson bracket together with $\{j_X(f), j_Y(h_Y)\} = 0$. If the interaction term vanishes for every f, then $dV(X_f, 0) = 0$. Conversely that annihilation condition makes every interaction bracket vanish. Lemma-style pointwise generation follows from nondegeneracy: at a chosen point, any tangent vector is the Hamiltonian vector of a smooth function after extending a local function by a bump function. Hence, when the full observable algebra is used, the condition is equivalent to V being locally constant along each connected X-fiber. □

Example 4.9 (Coupled oscillators revisited)**.** *For $V = kq_1q_2$, the interaction contribution to $\dot{p}_1$ is $\{p_1, V\} = -kq_2$, whereas its contribution to $\dot{q}_1$ is zero. The kinematical embeddings remain valid, but the momentum observable detects the coupling immediately. This calculation locates precisely which part of the factor dynamics fails rather than merely declaring the whole product "non-independent."*

5 Constraints and empirical realization

Assumption 5.1 (Regular constraint, A3)**.** *For a constraint map $C : X \to \mathbb{R}^k$, zero is a regular value whenever the regular-value theorem is invoked.*

Without A3, a constraint surface may be singular. Even under A3, the pullback of a symplectic form can be degenerate, and reduction may require a quotient whose smoothness must be established separately [1].

Proposition 5.2 (Interaction is extra structure, P4)**.** *The product symplectic structure and the embedded subsystem observables do not determine an interaction Hamiltonian.*

Proof. Both $V = 0$ and $V = (f \circ \pi_X)(g \circ \pi_Y)$ are smooth functions on the same product and leave its projections and symplectic form unchanged. If $dV \neq 0$, nondegeneracy makes their Hamiltonian vector fields different. □

Counterexample 5.3 (Singular constraint set)**.** *On $\mathbb{R}^2$ let $C(q,p) = q^2 - p^3$. The value zero is not regular at the origin because $dC_{(0,0)} = 0$. Its zero set has a cusp there and is not an embedded one-dimensional submanifold. Consequently the instruction "impose $C = 0$ and restrict the symplectic form" does not, without further analysis, produce a symplectic subsystem. This example shows why A3 is a genuine hypothesis rather than a harmless convention.*

Example 5.4 (Regular constraint with degenerate pullback)**.** *In $(\mathbb{R}^4, dq_1 \wedge dp_1 + dq_2 \wedge dp_2)$ impose $q_2 = 0$. This is a regular hypersurface. Its tangent vectors have $dq_2 = 0$, so the pulled-back two-form is $dq_1 \wedge dp_1$ and annihilates the nonzero tangent direction ∂_{p_2}. Regularity of the level set therefore does not imply that the restricted form is symplectic. A quotient by the characteristic direction may be appropriate, but its existence and physical interpretation are additional questions.*

Proposition 5.5 (Restriction of a process, P6)**.** *Let $F : X \to Y$ be a diffeomorphism and let constraints $C_X : X \to \mathbb{R}^k$ and $C_Y : Y \to \mathbb{R}^k$ satisfy $C_Y \circ F = C_X$. If zero is a regular value of both, then F restricts to a diffeomorphism $C_X^{-1}(0) \to C_Y^{-1}(0)$. If F is symplectic, the restricted map preserves the pulled-back two-forms, whether or not those forms are nondegenerate.*

Proof. The intertwining identity maps one zero set into the other, and the same identity for F^{-1} gives bijectivity. The regular-value theorem supplies embedded submanifolds, and restriction preserves smoothness in both directions. If i_X, i_Y are the inclusions, then $(F|)^* i_Y^* \omega_Y = i_X^* F^* \omega_Y = i_X^* \omega_X$. □

This proposition deliberately stops before reduction. When a pulled-back form has a characteristic distribution, obtaining a reduced symplectic space requires integrability, a well-behaved leaf space, and descent of the form. The cited treatments of symplectic reduction [1, 2] make such hypotheses explicit; none follows merely from categorical product.

Definition 5.6 (Empirical realization, D6)**.** *An empirical realization scheme consists of preparation procedures, measurement procedures, a partial representation map from preparations to states, prediction maps for measurements, and declared uncertainty bounds. These are mathematical data until connected to actual procedures and observations.*

Empirical statement 5.7 (Calibration input, E2)**.** *The assertion that a preparation represents a specified state, or that a meter implements a specified observable, is empirical input requiring calibration and uncertainty assessment. No property of the state manifold proves the correspondence.*

Counterexample 5.8 (Isomorphism does not fix empirical meaning)**.** *The map $F(q,p) = (p, -q)$ is symplectic on $\mathbb{R}^2$. If the first coordinate is calibrated as position in metres, applying F without transporting that calibration changes the meaning and dimensions of the readout. Mathematical isomorphism alone does not establish physical equivalence.*

New result claimed by this work 5.9 (Four-layer separation, N1)**.** *Within Definitions D1–D6, kinematical product, observable embedding, dynamical factorization, and empirical compositionality are logically distinct. None of the first three entails the fourth.*

Proof. Definition D5 supplies the first two layers. The coupled-oscillator counterexample supplies models with those layers but without dynamical factorization. Definition D6 adds realization data not determined by the mathematical system, while the calibration counterexample shows that an isomorphism can fail to preserve those data. Hence empirical compositionality is not entailed by the preceding layers. □

Conjecture 5.10 (Empirical compositionality, C2)**.** *A general empirical theory of composition may be organized by lax symmetric-monoidal realization maps from mathematical systems to operational preparation-and-measurement structures. The existence of a sufficiently general operational target category is not established here.*

6 Conclusion

The proved results establish a bounded formal claim: finite-dimensional classical systems admit categorical kinematical composition, and additive Hamiltonians yield factorized dynamics. Interactions, constraints, incomplete flows, and realization maps prevent promotion of this claim to a universal physical principle. The broader founding thesis remains an interpretation whose adequacy must be assessed against empirical practice.

References

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[2] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Springer, 1999. doi:10.1007/978-0-387-21792-5.

[3] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.

[4] J. Baez and M. Stay, "Physics, Topology, Logic and Computation," 2011. `https://arxiv.org/abs/0903.0340`.